

Auditor-Free Byzantine-Resilient Federated Learning via Functional Inner Products

Introduction

Federated Learning (FL) trains a shared model across many clients without exposing raw data, but remains vulnerable to **Byzantine model-poisoning attacks**, where adversarial clients manipulate gradients to disrupt convergence or induce targeted misclassification. This is amplified in **Privacy-Preserving FL (PPFL)**, where encrypted gradients block direct inspection.

Existing defenses rely on **auditor-based** architectures that cluster and score updates before aggregation. Auditors add:

- Significant computational and communication overhead,
- Cryptographic complexity and latency,
- A single-point-of-failure trust assumption.

Contributions

- An **auditor-free** Byzantine-resilient PPFL framework using lightweight **Functional Inner Product (FIP)** geometry on encrypted gradients.
- Two-stage geometric audit — *direction* then *magnitude* — with linear $\mathcal{O}(nd)$ complexity.
- State-of-the-art targeted accuracy (**97.69%**) on MNIST non-IID, surpassing the best auditor-based baseline without any external auditor.

Threat Model

- **Server:** honest-but-curious aggregator.
- **Adversarial clients:** up to **50%** Byzantine; label-flip (targeted) and scaled-magnitude (untargeted) attacks.
- **Attacker goal:** degrade accuracy or force targeted misclassification.
- **Defender goal:** detect malicious updates without decrypting individual gradients.

Methodology

The framework embeds Byzantine resilience directly into server-side aggregation — no external auditor needed. Each client i submits an encrypted gradient $E(g_i^{(t)})$; the server evaluates each update via its alignment with a trusted reference direction.

(1) **Functional Inner Product (FIP)**. Alignment of client i with the reference $g_{\text{ref}}^{(t)}$ is computed homomorphically:

$$s_i^{(t)} = \langle g_i^{(t)}, g_{\text{ref}}^{(t)} \rangle = \sum_{k=1}^d g_i^{(t,k)} g_{\text{ref}}^{(t,k)}$$

Only the **scalar** $s_i^{(t)}$ is decrypted; the gradient vector remains encrypted.

(2) **Reference Gradient (momentum-stabilized)**. The reference is anchored on a trusted bootstrap and updated smoothly:

$$g_{\text{ref}}^{(0)} = \nabla \mathcal{L}(w^{(0)}; D_{\text{init}}), \quad g_{\text{ref}}^{(t)} = \beta g_{\text{ref}}^{(t-1)} + (1 - \beta) g_{\text{global}}^{(t-1)}$$

with $\beta \in [0, 1]$ trading off history vs. new information, preventing adversarial drift.

(3) **Robust Statistical Bound**. Across alignment scores $X = \{s_i^{(t)}\}$, compute median μ and Median Absolute Deviation (MAD). The adaptive acceptance window is

$$\tau_{\min} = \mu \cdot \alpha, \quad \tau_{\max} = \mu + 5 \cdot \text{MAD}$$

where $\alpha \in [0, 1]$ adapts to the estimated attack rate $\rho^{(t)}$. **Accept** client i iff $\tau_{\min} \leq s_i^{(t)} \leq \tau_{\max}$.

(4) **Secure Aggregation**. Validated ciphertexts are summed under Paillier and decrypted only *after* aggregation:

$$E(g_{\text{global}}^{(t)}) = \sum_{i \in \mathcal{S}^{(t)}} E(g_i^{(t)})$$

Experimental Setup

- **Dataset:** MNIST, non-IID partitioning across clients
- **Model:** Logistic regression
- **Encryption:** Paillier additive homomorphic encryption
- **Attacks:** Targeted (label-flip) & untargeted (scaled); attack rates 10%, 20%, 30%, 50%
- **Metrics:** Target Accuracy (T_{acc}), Other-label Accuracy (O_{acc}), Overall Accuracy (ACC)

Privacy Guarantees

- Per-client gradients remain **encrypted end-to-end**; never decrypted at the server.
- Exactly **two decryptions per round**: per-client audit scalar and the final aggregate.
- Audit scalar reveals only the projection onto the trusted direction — not the gradient.
- Private key held only by the Trusted Key Authority.

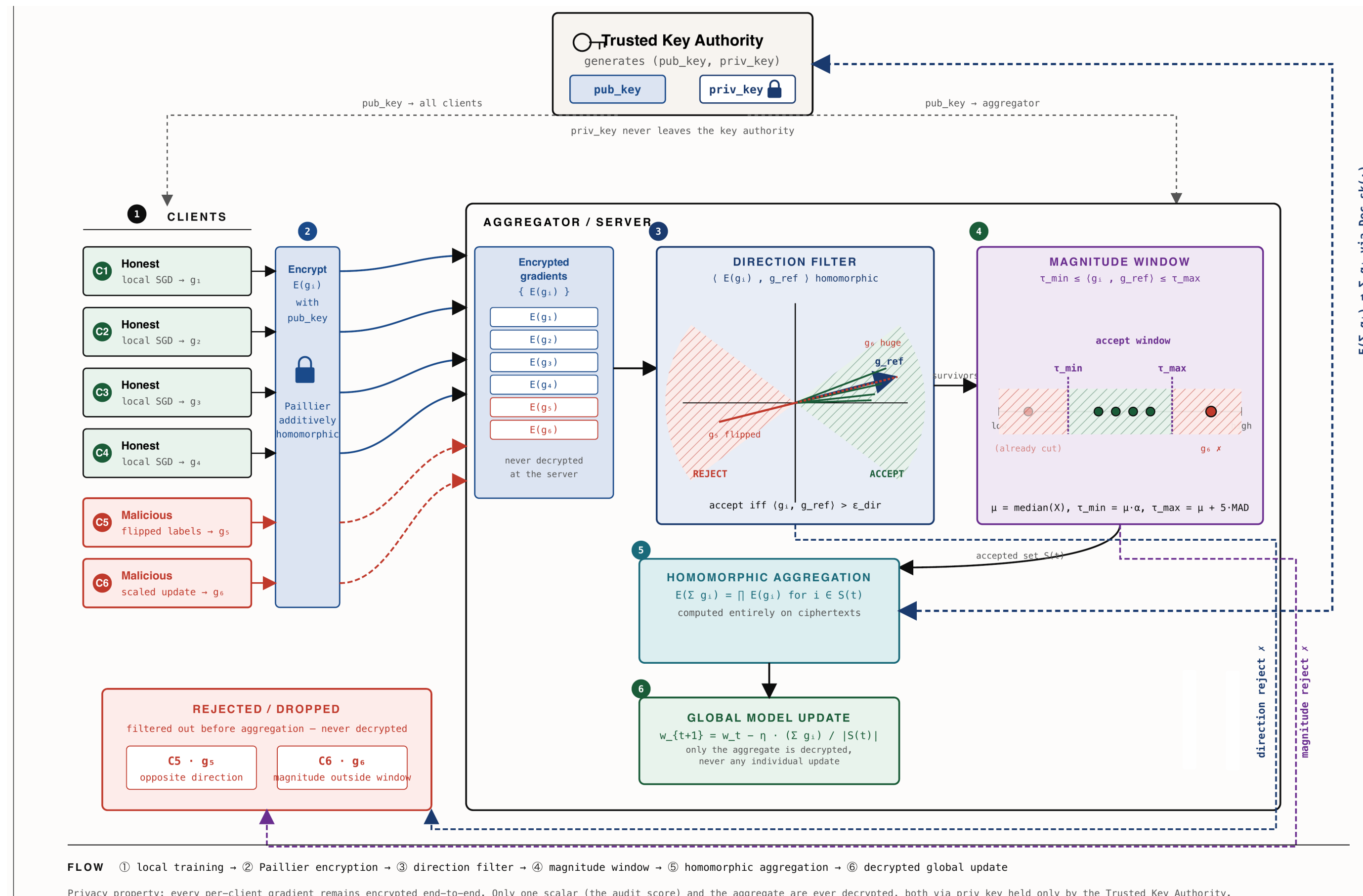
Complexity Comparison

Component	Auditor-Based [1]	Proposed
Clustering (GMM)	$\mathcal{O}(nd^2)$	—
Distance Computation	$\mathcal{O}(nd^2)$	—
Inner Product Audit	—	$\mathcal{O}(nd)$
Extra Comm. Hop	Yes	No
Scalability	Limited	Linear (n, d)

Key Highlights

- ✓ No external auditor needed
- ✓ **$\mathcal{O}(nd)$** complexity vs $\mathcal{O}(nd^2)$
- ✓ **97.69%** targeted accuracy (SOTA)
- ✓ Gradients stay encrypted end-to-end

System Architecture



Two-stage geometric audit on Paillier-encrypted gradients: (3) direction filter removes label-flip attacks; (4) magnitude window catches scaling attacks; (5) only validated ciphertexts are aggregated. Audit scalar and final aggregate are the **only** decryptions — individual gradients remain encrypted end-to-end.

Results – Defense Effectiveness

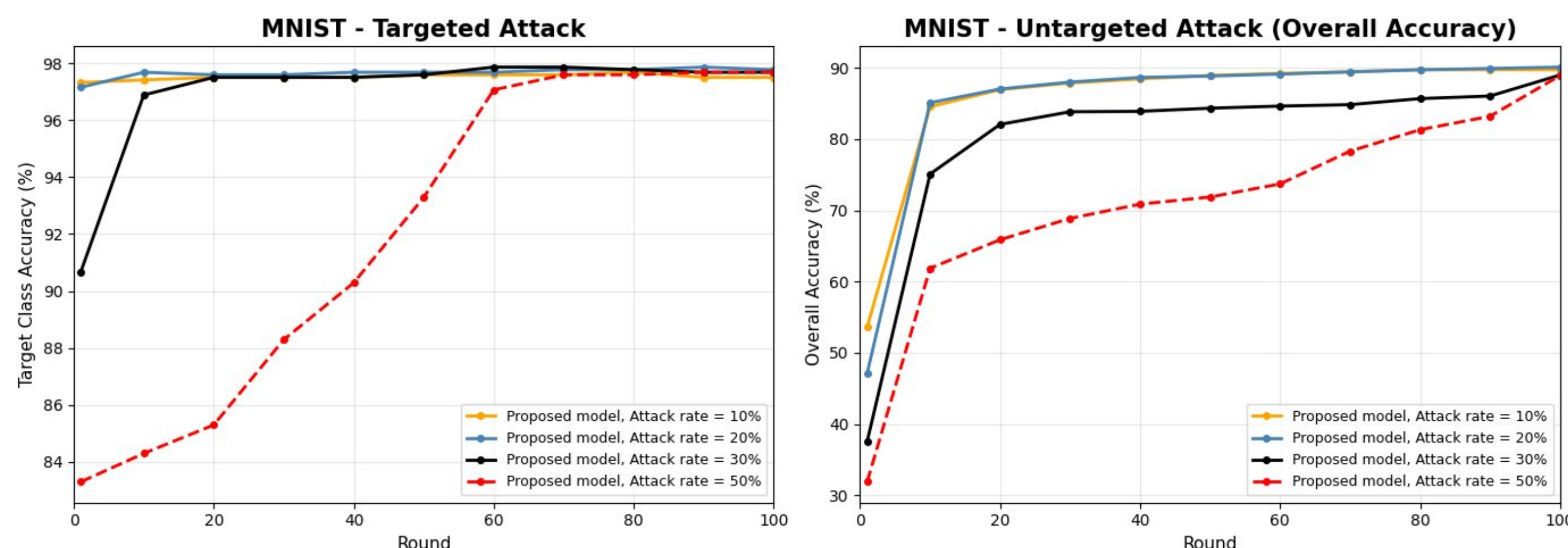
Setting	Without Defence			Proposed Defence		
	T_{acc}	O_{acc}	ACC	T_{acc}	O_{acc}	ACC
Non-IID, 50% attack	0.07	0.89	0.81	0.97	0.97	0.88

Targeted-attack mitigation: T_{acc} jumps from 0.07 $\rightarrow$ 0.97 without compromising other-class accuracy.

Comparison with Existing Methods

Defense	Targeted (%)	Untargeted (%)
Krum	71.4	78.3
Auror	32.5	49.9
PEFL	46.4	57.4
Sybils	89.7	89.5
FL-Trust	37.8	43.4
ShieldFL	90.1	89.4
Auditor-Based [1]	96.5	96.3
Proposed (FIP-PPFL)	97.69	88.82

Convergence Under Attack



MNIST non-IID across attack rates 10%–50%. Defense holds $\geq 97\%$ targeted accuracy through 30% attack rate; convergence preserved at 50%.

Conclusion & Future Work

This work presents an **auditor-free, Byzantine-resilient federated learning framework** that leverages functional inner product geometry and robust statistical thresholding to detect malicious updates in encrypted gradients. Experiments on MNIST show strong robustness against targeted and untargeted poisoning attacks while reducing computational and communication overhead. Future work will validate scalability on IID settings and broader datasets.

Resources

All code, scripts, and notebooks on GitHub:

Repository: github.com/guptadeepak2001/M.Tech_Thesis_2024pcs0024



References

- [1] A. Yazdinejad et al. *A robust privacy-preserving federated learning model against model poisoning attacks*. IEEE TIFS, vol. 19, pp. 6693–6708, 2024.
- [2] A. Zhou, A. B. Farimani. *Hamiltonian neural PDE solvers through functional approximation*. arXiv:2505.13275, 2025.